Cambridge (CIE) IGCSE Chemistry
Paper 2: Multiple Choice (Set D)
Extended

Saturday 26 April 2025

Morning (Time: 0 hours 45 minutes)

Total marks **/ 40**

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.

Materials

- A copy of the Periodic Table of elements is required for this paper.

Scan here to mark your mock exam
or visit the mock exams landing page for this course

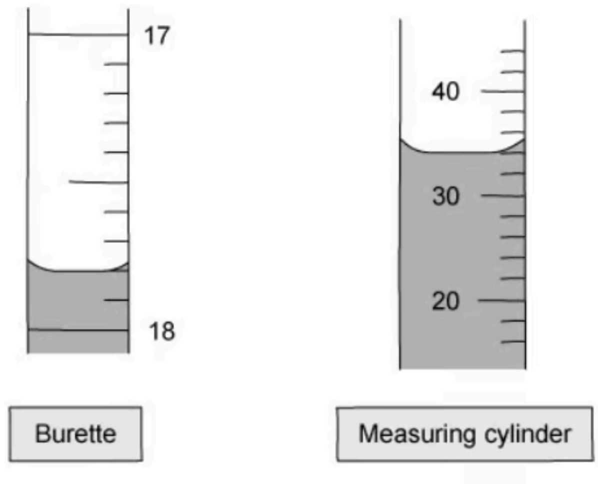
1 The diagram shows a cup of hot chocolate which was prepared by mixing boiling water with chocolate powder.

Which row correctly describes the particles of water in the air above the cup compared to those in the hot chocolate drink inside the cup?

	Particles move faster	Particles are closer together
A	✓	✓
B	✓	×
C	×	✓
D	×	×

(1 mark)

2 The diagram shows a burette and a measuring cylinder, both of which contain a liquid.



Which row shows the correct reading for both?

	Burette	Measuring cylinder
A	18.4	34
B	18.4	56
C	17.8	34
D	17.8	56

(1 mark)

3 Element X contains 12 of each sub-atomic particle.

What is the correct Group number for element X?

A. Group 0

B. Group I

C. Group VII

D. Group II

(1 mark)

4 Solid P melts at exactly 60^{o}C and boils at exactly 310^{o}C. Solid Q dissolves in ethanol and shows a green colour and a yellow colour when analysed using paper chromatography.

Which row is correct?

	Contains only one substance	**Contains more than one substance**
A	Q	P
B	Q	-
C	-	Q
D	P	Q

(1 mark)

5 Statements 1, 2 and 3 are about diamond and graphite.

1. They have atoms that form three equally strong bonds
2. They are both formed from the same element
3. They each conduct electricity

Which statement(s) is / are correct?

A. 1 only

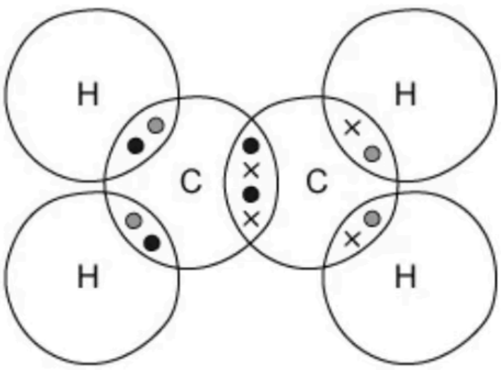
B. 2 only

C. 1 and 3

D. 1 and 2

(1 mark)

6 A student has drawn a dot and cross diagram to represent a molecule of ethene, C_2H_4, as shown below.



What is wrong with the students drawing?

A. The number of electrons shared between the carbon atoms is incorrect.

B. The number of electrons shared between each carbon and hydrogen is incorrect.

C. There is nothing wrong with the drawing.

D. The total number of electrons shared is incorrect.

(1 mark)

7 What is the balanced equation for the reaction of methane and oxygen?

A. $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + H_2O(g)$

B. $CH_4(g) + O_2(g) \rightarrow CO_2(g) + 2\ H_2O(g)$

C. $2\ CH_4(g) + 4\ O_2(g) \rightarrow 2\ CO_2(g) + H_2O(g)$

D. $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(g)$

(1 mark)

8 A compound contains two atoms of oxygen and one atom of calcium. It also contains hydrogen in sufficient amounts so that the overall charge of the molecule is neutral.

What is the molecular formula for the compound?

A. $Ca(OH)_4$

B. $Ca(OH)_2$

C. H_4CaO_2

D. CaO_2H_2

(1 mark)

9 Aluminium and oxygen react to produce aluminium oxide.

$$4\ Al + 3\ O_2 \rightarrow 2\ Al_2O_3$$

How many kilograms of aluminium oxide are formed when 5.4 kg of aluminium react with excess oxygen?

A. 5.4 kg

B. 88 kg

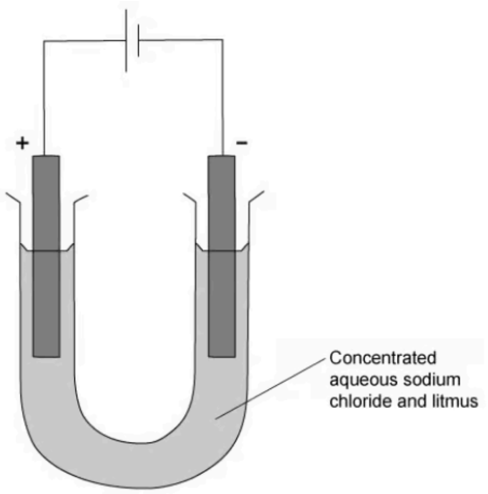
C. 10.2 kg

D. 33.6 kg

(1 mark)

10 Electricity is passed through concentrated aqueous sodium chloride. Inert electrodes are

used.



Which products are formed at the electrodes?

	cathode	anode
A	hydrogen	chlorine
B	hydrogen	oxygen
C	sodium	chlorine
D	sodium	oxygen

(1 mark)

11 What is the correct balanced symbol equation for the overall reaction in a hydrogen-oxygen fuel cell?

A. $2H_2 \rightarrow 4H^+ + 4e^-$

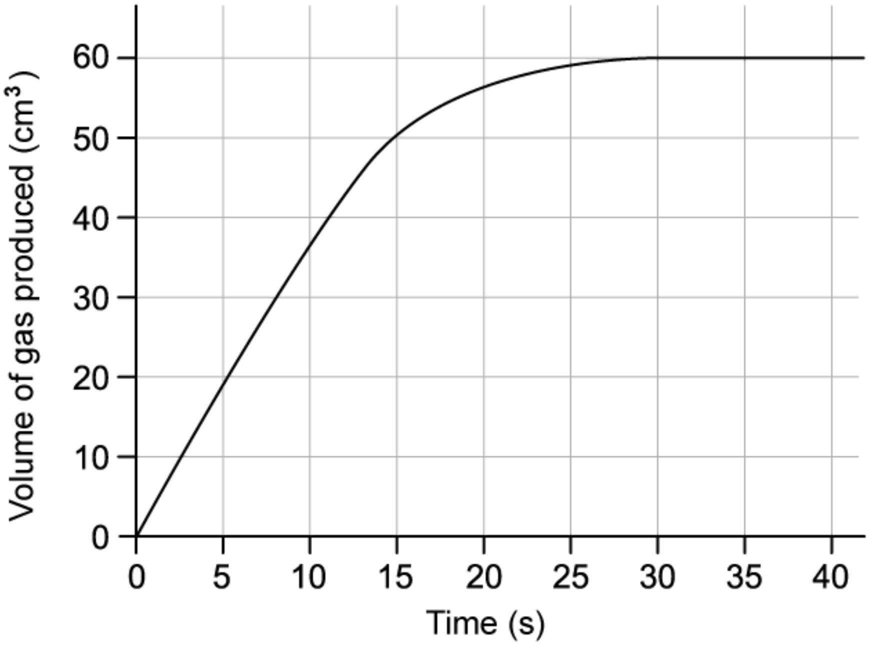
B. $4H^+ + O_2 + 4e^- \rightarrow 2H_2O$

C. $2H_2 + O_2 \rightarrow 2H_2O$

D. $2H_2O \rightarrow 2H_2 + O_2$

(1 mark)

12 The rate of a reaction was monitored by recording the volume of gas produced every 5 seconds.



What is the mean rate of reaction in the first 10 seconds of the reaction, in cm^3/s?

A. 2.9

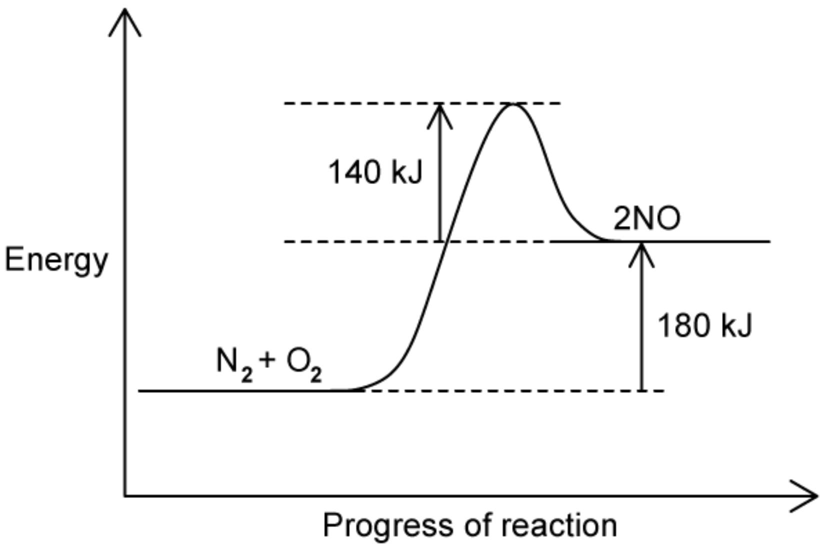
B. 3.0

C. 3.5

D. 2.5

(1 mark)

13 The reaction profile for making nitrogen monoxide, NO, is shown below:



What is the activation energy for this reaction?

A. +140 kJ

B. +180 kJ

C. +320 kJ

D. -330 kJ

(1 mark)

14 A student is given a white insoluble solid to analyse. He first added hydrochloric acid and observed bubbles of colourless gas. He then added a small amount of aqueous ammonia to the solid-acid mixture and observed a white precipitate being formed. The precipitate dissolved in excess aqueous ammonia.

What is the identity of the white solid?

A. Ammonium nitrate

B. Aluminium nitrate

C. Zinc carbonate

D. Calcium carbonate

(1 mark)

15 Samples of two solids, P & Q, are separately dissolved in water and made into solutions.

Which combination of P & Q solutions, would give an insoluble product on mixing?

	P	**Q**
A	potassium chloride	sodium hydroxide
B	ammonium chloride	potassium sulfate
C	sodium nitrate	magnesium sulfate
D	barium chloride	sodium sulfate

(1 mark)

16 Properties of four alkali metals labelled are shown in the table. Which one is placed lowest in the group?

	melting point / ^oC	**density / g cm^{-3}**
A	39	1.50
B	28.5	1.93
C	98	0.98
D	63.5	0.86

(1 mark)

17 Which of the following properties is not indicative of a base?

A. Carbon dioxide is released when it reacts with a carbonate.

B. A salt is formed on reaction with an acid.

C. Ammonia is released on reaction with an ammonium salt.

D. Universal indicator paper turns blue when placed in an alkaline solution.

(1 mark)

18 Ionic equations 1 and 2 show how substance **X** behaves in aqueous solution.

1	$X^- + H^+ \rightarrow XH$
2	$X \rightarrow X^- + H^+$

Which row correctly describes substance X?

	acid	**base**
A	1	1
B	1	2
C	2	2
D	2	1

(1 mark)

19 Copper(II) sulfate can be prepared by reacting solid copper(II) carbonate with hot sulfuric acid. The copper(II) carbonate is always added in excess.

Why is the solid reactant added in excess?

A. To ensure all of the acid reacts.

B. To make the reaction go faster.

C. To ensure all of the solid reactant has reacted.

D. To increase the product yield.

(1 mark)

20 Potassium and krypton are elements in Period 4. Potassium is at the beginning of the period and is more metallic than krypton which is at the end of the period.

What makes potassium more metallic than krypton?

A. It has less electrons than krypton.

B. It has less protons than krypton.

C. It has less electron shells.

D. It has less outer shell electrons.

(1 mark)

21 An element has an electronic configuration of 2,8,6.

Which group and period can the element be found in?

	group	period
A	6	3
B	6	8
C	3	6
D	8	2

(1 mark)

22 Which of the following observations does not apply to Group I elements?

A. They are relatively soft metals.

B. They corrode quickly when exposed to air.

C. They react very quickly with water releasing hydrogen gas

D. They produce acidic solutions in reaction with water.

(1 mark)

23 Which reaction does **not** occur in the extraction of aluminium?

A. $Al^{3+} + 3\ e^{-} \rightarrow Al$

B. $2\ Al_2O_3 + 3\ C \rightarrow 4\ Al + 3\ CO_2$

C. $2\ O^{2-} \rightarrow O_2 + 4\ e^{-}$

D. $C + O_2 \rightarrow CO_2$

(1 mark)

24 Calcium, copper, iron and magnesium are metals. They can be placed in order of reactivity.

Which of the following statements about these metals is correct?

A. Iron reacts with steam but magnesium does not.

B. Copper reacts with dilute hydrochloric acid to form copper (II) chloride.

C. Magnesium and calcium both react with hot water.

D. Iron (II) oxide cannot be reduced by heating strongly with carbon.

(1 mark)

25 The transition elements are used extensively as catalysts. Which statement explains why they are such good catalysts?

A. Their oxidation state does not change.

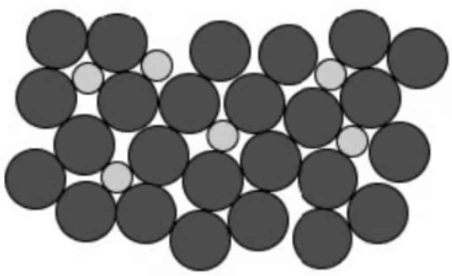
B. They have a large surface area.

C. They can have variable oxidation states.

D. They form coloured compounds.

(1 mark)

26 The structure of an alloy is shown below. Alloys are mixtures of metals with other metals or non-metals.



They are usually much stronger and harder than pure metals. Which statement explains these features?

A. The irregular sizes of the atoms distorts the regular arrangement in the metal structure, meaning more force is required to bend the metal

B. The irregular sizes of the atoms distorts the regular arrangement in the metal structure as the bonding is disrupted when the atoms are added

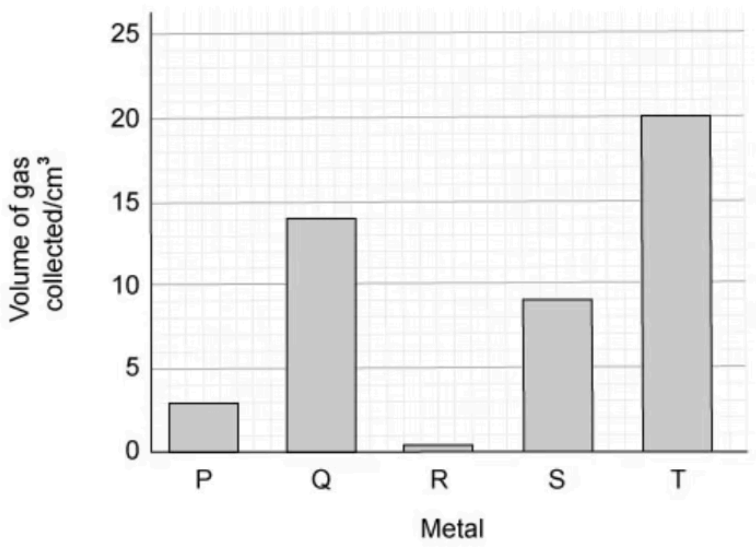
C. The regular sizes of the atoms distorts the irregular arrangement in the metal structure, meaning more force is required to bend the metal

D. A mixture of metals is tougher than a pure metal because a mixture has more intermolecular forces than a pure metal.

(1 mark)

27 Five different metals **P**, **Q**, **R**, **S** and **T** were reacted with dilute hydrochloric acid.

The volume of hydrogen gas produced after one minute was measured. The results are shown in the bar chart below:



What is the order of reactivity of the metals?

	most reactive	→	→	→	least reactive
A	P	Q	R	S	T
B	R	P	S	Q	T
C	T	Q	S	P	R
D	T	S	R	Q	P

A. Option A

B. Option B

C. Option C

D. Option D

(1 mark)

28 Aqueous sodium hydroxide is used in chemical analysis.

Which statement about aqueous NaOH is correct?

A. A white precipitate is formed when added to a solution containing sulfate ions

B. A green precipitate is formed upon addition to a solution of iron(II) ions which does not dissolve in excess

C. A green precipitate is formed upon addition to a solution of iron(II) ions which dissolves in excess

D. An acidic gas is produced when added to ammonium chloride

(1 mark)

29 Which pair of substances would make an NPK fertiliser?

A. ammonium phosphate and potassium chloride

B. calcium phosphate and potassium chloride

C. ammonium sulfate and potassium nitrate

D. potassium nitrate and calcium carbonate

(1 mark)

30 The production of sulfur trioxide during Stage 2 of the Contact process involves a reversible reaction. Which row describes the conditions of temperature and pressure and the percentage yield of sulfur trioxide?

	temperature / °C	**pressure / atm**	**percentage yield**
A	150	1 - 2	70
B	450	1 - 2	98
C	450	200	98
D	150	200	70

(1 mark)

31 Which is the correct reason for carrying out each step when treating water?

	Adding carbon	Chlorination	Sedimentation
A	kill microbes	remove salts	remove solids
B	remove salts	remove odours	kill microbes
C	remove odours	kill microbes	remove solids
D	remove odours	remove solids	remove salts

(1 mark)

32 Carbon monoxide and steam react together exothermically to produce the alcohol methanol, as shown in the equation.

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

Both reactants and the product are gaseous. Which conditions of pressure and temperature would provide the highest yield of CH_3OH?

A. **pressure -**low
temperature -low

B. **pressure -**low
temperature -high

C. **pressure -**high
temperature -low

D. **pressure -**high
temperature -high

(1 mark)

33 The results of different chemical tests on a solution of compound **Y** are shown.

Test	Result
Acidified barium nitrate added	White precipitate formed
Aqueous sodium hydroxide added	White precipitate formed, soluble in excess
Aqueous ammonia added	White precipitate formed, insoluble in excess

What is compound **Y**?

A. Aluminium sulfate

B. Aluminium chloride

C. Zinc sulfate

D. Zinc chloride

(1 mark)

34 Which statement about the extraction of iron from its ore is **not** correct?

A. Iron can be extracted by reduction with carbon

B. Iron is easier to extract than zinc

C. Iron is easier to extract than copper

D. Iron is extracted from the ore, hematite

(1 mark)

35 Hydrocarbons separated by fractional distillation of petroleum can be cracked to make useful shorter chain products.

Which substance is not a product of the cracking of propane, M_r 44?

A. C_4H_8

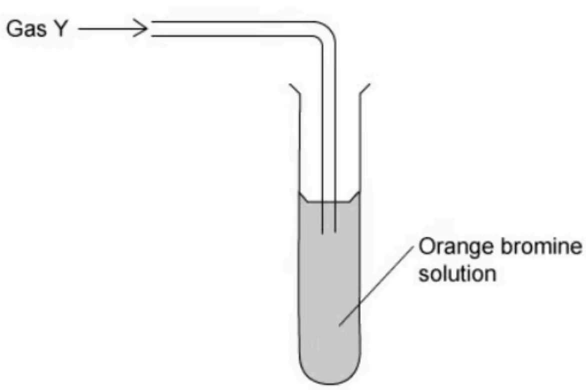
B. C_2H_4

C. C_3H_6

D. H_2

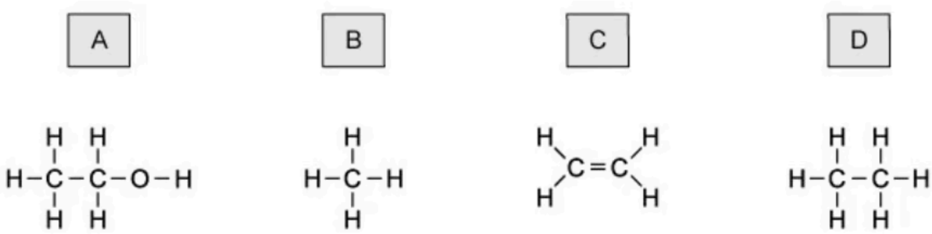
(1 mark)

36 The apparatus below was used to test gas Y



The bromine solution quickly went colourless.

What is the correct structure of gas Y?



(1 mark)

37 The following reaction takes place in the presence of light.

$$CH_4 + Cl_2 + \text{light} \rightarrow CH_3Cl + HCl$$

What is the name for this reaction?

A. Combustion

B. Polymerisation

C. Thermal decomposition

D. Substitution

(1 mark)

38 Which of these is an advantage of producing ethanol by fermentation of sugar compared to the catalytic addition of steam to ethene?

A. The process uses high temperatures.

B. The alcohol produced is purer.

C. The process is faster.

D. The process uses renewable raw materials.

(1 mark)

39 Which of the equations show the incomplete combustion of methane?

A. $2\ CH_4 + 4O_2 \rightarrow 2\ CO_2 + 4\ H_2O$

B. $CH_4 + O_2 \rightarrow C + 2\ H_2O$

C. $CH_4 + O_2 \rightarrow C + 2\ H_2$

D. $2\ CH_4 + 4\ O_2 \rightarrow 2\ CO_2 + 4\ H_2$

(1 mark)

40 In which row are the monomer and polymer chain correctly matched?

	monomer	part of polymer chain
A	$CH_3CH{=}CHCH_3$	$–CH(CH_3)–CH(CH_3)–CH(CH_3)–CH(CH_3)–$
B	$CH_2{=}CHCl$	$–CHC–CHCl–CHCl–CHCl –$
C	$CH_3CH{=}CH_2$	$–CH_3–CH–CH_2–CH_3–CH–CH_2–$
D	$CH_2{=}CHCH_2CH_3$	$–CH_2–CH_2–CH_2–CH(CH_2CH_3)–$

(1 mark)